

Changes in intestinal microbiota and their effects on allogeneic stem cell transplantation

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Abstract

The human intestinal microbiota is essential for microbial homeostasis, regulation of metabolism, and intestinal immune tolerance. Rapidly evolving understanding of the importance of the microbiota implicates changes in the composition and function of intestinal microbial communities in an assortment of systemic conditions. Complications following allogeneic stem cell transplant now join the ever-expanding list of pathologic states regulated by intestinal microbiota. Dysbiosis, or disruption of the normal ecology of this microbiome, has been directly implicated in the pathogenesis of entities such as *Clostridium difficile* infections, graft-versus-host disease (GVHD), and most recently disease relapse, all of which are major causes of morbidity and mortality in patients undergoing allogeneic stem cell transplant. In this review, we elucidate the key origins of microbiotic alterations and discuss how dysbiosis influences complications following allogeneic stem cell transplant. Our emerging understanding of the importance of a balanced and diverse intestinal microbiota is prompting investigation into the appropriate treatment of dysbiosis, reliable and early detection of such, and ultimately its prevention in patients to improve the outcome following allogeneic hematopoietic stem cell transplant.

1 | INTRODUCTION

The human body is colonized by thousands of different microbial species that are key to our survival.¹ Over the past 15 years, the understanding of human microbiome has increased exponentially. Early identification of microbes, primarily of the gut, was studied by means of cultures and characterization of physiologic properties. Newer methods involving the use of mass deoxyribonucleic acid sequencing and fluorescence in situ hybridization targeting the 16S ribosomal ribonucleic acid² have allowed a more comprehensive evaluation of the microbiome.

The intestinal microbiota is essential in maintaining and facilitating human health and well-being. Numerous microbiotic mechanisms have been identified which demonstrate a symbiosis between human wellness and intestinal commensals, most notably a substantial influence on adaptive immunity.³ In contrast, dysbiosis, defined as perturbation that disrupts the balanced microbiome, may prolong, exacerbate or induce detrimental health effects.⁴ This has most strongly been documented in inflammatory bowel disease where there are increases in *Proteobacteria*, *Firmicutes bacillus*, and *Actinobacteria*. Fecal microbiota transplant (FMT) that restores normal intestinal microbiota and results in long lasting remission in these patients provides additional support

for this notion.⁵ Dysbiosis has also been implicated in the pathophysiology of colorectal cancer, multiple sclerosis, diabetes mellitus, obesity, and allergies.^{6–10}

Recipients of allogeneic hematopoietic stem cell transplant (HSCT) are at very high risks for many infectious and inflammatory complications, because of their underlying malignancies, and extensive exposure to systemic antibiotics and chemotherapeutic agents that render them particularly vulnerable to major shifts in the intestinal microbiome. An understanding of these microbiologic shifts and how they may influence complications following allogeneic HSCT is crucial in devising strategies to prevent and treat transplant-related complications in this unique population.

2 | INTESTINAL DYSBIOSIS AND ALLOGENEIC HSCT

2.1 | Intestinal microbiota and GVHD

Early investigations into the role of microbiota following allogeneic HSCT demonstrated a significant decrease in mortality and acute GVHD in mice transplanted in a germ-free environment.^{11,12} A decade later a human trial involving patients who underwent allogeneic HSCT

showed similar benefits of intestinal decontamination and infection prophylaxis in isolated environment.¹³ This led to the practice of instituting strict isolation and an antimicrobial decontamination regimen for these patients. However, this practice fell out of favor within a few years when the initial human data could not be consistently replicated.^{14,15} The conflicting results of these studies were most likely due to the fact that, although intestinal decontamination was practiced in the later studies, it was carried out in an unprotected clinical setting instead of microbiologically isolated environment.¹⁴ Systemic infections were, therefore, more common. Furthermore, there were differences in the intensity and types of chemoradiation given to the patients in these studies.^{13–15} More recently rifaximin, when compared to the standard combination ciprofloxacin and metronidazole, resulted in lower fecal enterococcal positivity and ultimate mortality benefit.¹⁶ These conflicting data, therefore, highlight the complex interactions between intestinal microbiota, intestinal inflammatory process induced by chemoradiation given to the patients prior to the allogeneic HSCT, and systemic infections.

A head-to-head comparison of metronidazole and ciprofloxacin with ciprofloxacin alone bolstered the realization that alteration of the intestinal microbiota has a significant role in the development of GVHD.¹⁷ However, further effort to dissect the intestinal microbiota was hampered by a lack of reliable techniques until the recent availability of the next generation sequencer for mass genomic sequencing. Using this molecular technique, it was found that, although the intestinal flora diversity in patients with hematologic malignancy prior to allogeneic HSCT was similar to that in the healthy population,^{18,19} a loss of the diversity occurred after allogeneic HSCT, even before the development of any GVHD. Holler et al¹⁹ found that there was 21% increase in the prevalence of *Enterococcus*, especially *E faecium*, and a corresponding drop in *Firmicutes* and other commensal phyla. The restriction of diversity became even more prominent among patients who developed GVHD, and a restricted diversity of the intestinal microbiome was found to be an independent risk factor for increased mortality after allogeneic HSCT.^{18,20} Jenq et al¹⁸ showed in a murine model that intestinal inflammation, induced by high dose chemoradiation given as the transplant preparative regimen, led to shifts in the microbiome, specifically expansion of *Lactobacillales* and loss of *Clostridiales* (and other *Firmicutes* species), supporting the effects of inflammation in modifying the intestinal microbiota. These changes were amplified further in mice that developed GVHD. This is not surprising since the development of intestinal GVHD intensifies the ongoing intestinal inflammatory process induced by the transplant preparatory regimens. The expansion of *Lactobacillales* appeared to be compensatory to modulate intestinal GVHD, rather than being the causative factor because mice depleted of intestinal *Lactobacillales* still developed GVHD. The GVHD in the *Lactobacillales*-depleted mice was more severe and resulted in poorer survival of the mice. Repletion of the intestinal *Lactobacillales* improved the transplant outcome. Therefore, the interaction between the *Lactobacillales* and intestinal inflammation appeared to be bidirectional.

Furthermore human studies of the loss of diversity of the intestinal microbiome after allogeneic HSCT found the correlation between

reduced *Blautia* genus and a decreased incidence of acute GVHD and overall mortality.²¹ Expansion of *Enterococci* has also been shown to predict an increased risk of GVHD,¹⁹ although a subsequent study could not replicate the finding.²¹

2.2 | Interactions between intestinal microbiota and inflammatory cytokines during allogeneic HSCT

Various laboratory observations^{18,19} highlight the complex and multidirectional interplay between inflammation, microbiota, and immune reactivity in the intestine following allogeneic HSCT (Figure 1). Mucosal damage induced by high dose chemoradiation given to the patients prior to the transplant sets off a cascade of inflammatory cytokine release, including of tumor necrosis factor (TNF), type I interferons, interleukin (IL)-1, and IL-6.²² These cytokines not only directly modify intestinal microbiota composition but also upregulate the expression of major histocompatibility complex and immune costimulatory molecules, creating the optimal conditions for the development of GVHD. The increased intestinal permeability induced by tissue damage allows the translocation of endotoxins across the intestinal mucosa, amplifying further the activation of host's innate immune system and cytokine release.²³ A whole host of damage-associated molecular patterns (DAMPs) released from damaged cells and pathogen-associated molecular patterns (PAMPs) by the microbiota provide additional initiation pathways for this cascade of activation.²⁴

Intestinal crypt damage induced by the transplant preparative regimens affects the production of Paneth cell antimicrobial peptides, particularly of Reg3alpha that is the main peptide for limiting Enterococcal overgrowth.²⁵ Paneth cell antimicrobial peptides are responsible for maintaining a healthy and intact intestinal mucosa. Coupled with a loss of colonization resistance exerted by commensal bacterial suppressed by metronidazole, lowering of the Paneth cell antimicrobial peptides not only exacerbates the intestinal inflammation induced by the high dose chemoradiation given to the patients prior to allogeneic HSCT but also renders the host more susceptible to intestinally originated infections. This may explain the finding that, although pretreatment with the combination ciprofloxacin and metronidazole induced a 10-fold drop in anaerobic bacterial culture growth following allogeneic HSCT, it is associated with an increase in *Enterococcus*.¹⁷ *Enterococcus* has been shown to have a directly detrimental effect on colonic epithelial tissues through toxin-mediated gut inflammation and damage as well as induction of macrophage TNF alpha release,^{26,27} and ultimately contribute toward the development of GVHD. The end impact on intestinal damage and GVHD development due to changes in PAMPs induced by chemoradiation and use of antimicrobials is highly determined by which of the two is the more dominant player in the interaction. Antimicrobial-induced depletion of intestinal microbiota has also been shown, at least in mice, to be associated with changes in the serotonin and tryptophan hydroxylase 1 that modify the intestinal motility.²⁸ Once GVHD develops, it sets off additional cycles of cytokine release and T-cell activation. Both these processes not only induce additional mucosal damage but also amplify the major shift in intestinal microbiota.

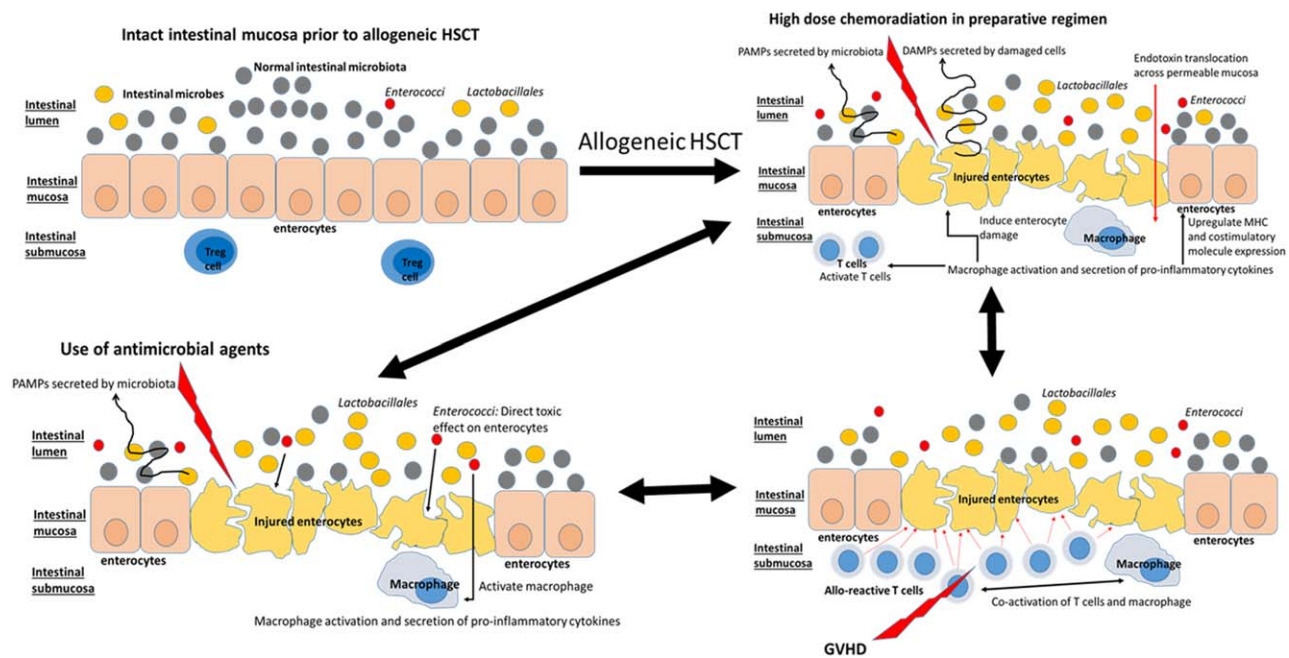


FIGURE 1 The multidirectional interactions between intestinal microbiota, GVHD, and mucosal damage [Color figure can be viewed at wileyonlinelibrary.com]

The multidirectional impact of cytokines, GVHD, and intestinal microbiome is now an area of great interest to physicians involved in allogeneic HSCT. A study is currently ongoing to compare the fecal microbiotic differences between those who develop chronic GVHD following allo-HSCT and those who do not (NCT02398708). This study will measure 17 cytokines in serum and stool in these individuals. It is likely that patients with chronic GVHD will exhibit a specific cytokine profile, and stool analysis in these patients could provide an initial foray into a possible mechanism into the interaction between the intestinal microbiota and T cells responsible for chronic GVHD.

2.3 | Diet, intestinal microbiota, and GVHD

Diet plays a major role in determining the composition of the intestinal microbiome.²⁹ It, therefore, follows that certain diet may contribute to the development of GVHD. Enteral feeding has been shown in several studies to protect against GVHD.³⁰ Mice and human subjects receiving total parenteral nutrition exhibited increases in interferon gamma and reduction in IL-4, IL-10, gut CD4⁺ cells,^{31,32} a pattern of immune responses that are proinflammatory. The progressive move away from the routine use of parenteral feeding in patients undergoing allogeneic HSCT could also explain the discrepancies of results in various studies of intestinal decontamination following allogeneic HSCT.^{13–15}

The beneficial effects of enteral feeding may be related to the prebiotic properties of fiber from select fruits and vegetables.³³ Prebiotics are indigestible carbohydrate that is fermented by colonic flora or commensals to produce short chain fatty acids (SCFAs) such as butyrate, acetate, and propionate that are utilized as an energy source by and promote the health of the human enterocyte.³⁴ Some of these SCFAs are pivotal in anti-inflammatory pathways through the induction of regulatory T-cells which produce anti-inflammatory cytokines (eg, IL-10)

that may modulate the incidence and severity of GVHD (Figure 2).^{35–37} Butyrate, a SCFA, promotes intestinal barrier function, increases trans-epithelial electrical resistance and decreases inulin permeability.^{38,39} Reduced levels of butyrate occurred in mucosal tissue following allogeneic HSCT and were associated with decreased histone acetylation and increased enterocyte apoptosis.³⁹ Restoration with local administration of exogenous butyrate restored enterocyte junctional integrity, reduced apoptosis and mitigated GVHD. Similar results were also obtained with the introduction of high butyrate-producing *Clostridia*.

Additional intestinal-protective effect may be conferred by the indole metabolites produced by intestinal commensal bacteria. Restoration or preservation of intestinal commensal bacteria by probiotics ensures the continued production of the indole metabolites that have been found in mice to modulate incretin secretion from colonic L cells⁴⁰ and increases epithelial tight-junction resistance.⁴¹

A study showed that enteral supplementation with fiber and glutamine, in addition to oligosaccharides, conferred a mortality benefit and a trend toward lesser *Enterococcus* translocation in patients undergoing allogeneic HSCT.⁴² Of note, however, all the patients in this study also received a probiotic formulation which might also have directly altered the intestinal microbiome and thus perhaps its direct and indirect effects on mucosal integrity.

The effect of probiotics on the intestinal microbiome, mucosa, and on re-establishing favorable commensal colonization following allogeneic HSCT has also been investigated. There exists extensive variability in the preparations available, including single or multistrain products. Despite a lack of U.S. Food and Drug Administration (FDA) regulation, the most well-studied with regards to safety are *Lactobacillus*, *Bifidobacterium*, and *Saccharomyces boulardii*. Certain organisms in probiotic preparations have been found to mediate an immunoprotective effect on the intestinal mucosa. In vitro data on the effect of probiotics on

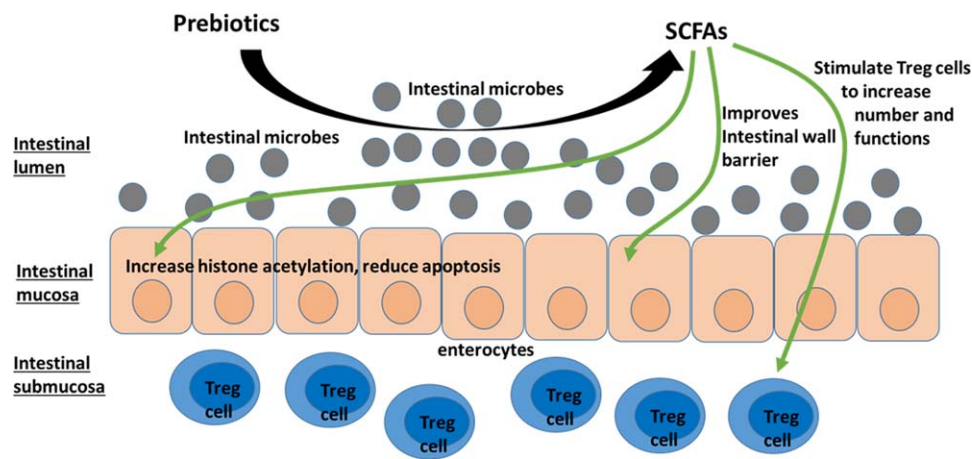


FIGURE 2 Mechanisms of action of prebiotics and SCFAs in modulating mucosal damage [Color figure can be viewed at wileyonlinelibrary.com]

human intestinal tissue have shown immunomodulation of intestinal tissue dendritic cells to produce IL-10, decrease interferon gamma release by intestinal T-cells and decreased lipopolysaccharide (LPS)-induced production of the proinflammatory IL-12.⁴³ A murine study showed a protective effect of the probiotics on the development of GVHD and improved overall mortality when given a prominent species of *Lactobacillus*.⁴⁴

Human studies investigating the effect of probiotics in patients undergoing allogeneic detailing probiotic fungemia and bacteremia in patients that were taking or given probiotic formulations, including patients who were immunocompromised by processes other than myeloablation and/or immunosuppressive medications (eg, human immunovirus, critical illness).^{45,46} However, a study involving 31 pediatric and adolescent undergoing allogeneic HSCT patients found that administration of *Lactobacillus* day-7 during the conditioning regimen through day +14 did not result in *Lactobacillus* bacteremia, although the study also did not find any significant difference in the incidence of GVHD.⁴⁷ Another study is currently ongoing on the use of the probiotic *L. rhamnosus* GG in patients who have attained a hematologic recovery following allogeneic HSCT to evaluate the incidence of both acute and chronic GVHD in conjunction with fecal analysis of the microbiome, duration of immunosuppressive therapy, and infection as well as measurement of numerous inflammatory mediators (eg, IL-2, IL-6, IL-8, IL-10, TNF-alpha, conventional T cells, regulatory T cells, B cells, and natural killer cells) and known biomarkers associated with GVHD (NCT02144701).

3 | INTESTINAL DYSBIOSIS AND *CLOSTRIDIUM DIFFICILE* INFECTION FOLLOWING ALLOGENEIC HSCT

A discussion of the effects of a disrupted intestinal microbiome on the development of GVHD following allogeneic HSCT is not complete without including *C. difficile* infection (CDI) since both diseases are related to intestinal dysbiosis and the presence of one affects the development of the other. CDI is one of the most commonly observed complications following allogeneic HSCT, estimated to occur in 10% to 15% of patients.^{48–52} With the emergence of the more virulent NAP-

1/027 strain, the incidence of CDI is expected to increase rapidly. CDI is often a sequela of broad-spectrum antibiotics which have considerable activity against protective commensals, leading to large shifts in the intestinal microbiome.⁵³

Asymptomatic patients with hematologic malignancies prior to allogeneic HSCT have been shown to have a *C. difficile* colonization rate ranging from 8% to 29%, with approximately 12% colonized with a toxigenic strain.^{51,54} In contrast, *C. difficile* colonization rate is much lower in the general population^{55,56} and only approximately 8% colonized with the toxigenic strain.⁵⁷

CDI in patients who underwent allogeneic HSCT was not associated with any increased risk of mortality.⁵² There are several reported risk factors for the incidence of CDI following allogeneic HSCT, including carbapenem use,⁵⁸ previous *C. difficile* colonization,⁵¹ CDI within nine months before allogeneic HSCT,⁵² total body irradiation, cord blood transplant, and onset of acute GVHD up to one year after allogeneic HSCT.⁴⁸ Matched-related donor transplants were associated with a lower incidence of peri-transplant CDI.⁵² In this study, CDI was found to occur much earlier and in nearly 40% of patients if there was confirmed CDI within the preceding nine months prior to allogeneic HSCT,⁵² raising the question of whether these patients might benefit from prophylactic metronidazole or oral vancomycin during the high-risk periods early after allogeneic HSCT.

Based on the multidirectional interactions between intestinal microbiota, mucosal inflammation, and GVHD (Figure 1), the association between CDI and the development of intestinal GVHD in patients undergoing allogeneic HSCT^{59,60} is not surprising. CDI induces local inflammatory reactions, causing the release of DAMPs and translocation of endotoxins across the intestinal mucosa. As a result of cytokine release, major histocompatibility complex (MHC) expression and immune costimulatory molecules are upregulated and allo-reactive T cells activated. These events provide the platform for the development of intestinal GVHD. Patients with CDI, even when the disease was mild, were more likely to develop GVHD and die from the disease.⁵⁹

Since CDI occurs due to dysbiosis, it follows that probiotics may have a role in preventing CDI in this group of patients. Human studies investigating the effect of probiotics in CDI have remained

controversial, although meta-analyses have provided some evidence in favor of their use in those specific clinical circumstances to prevent CDI.^{61,62}

4 | INTESTINAL DYSBIOSIS AND DISEASE RELAPSE FOLLOWING ALLOGENEIC HSCT

If intestinal microbiota interacts with and regulates the host immune system and modulates the development of GVHD, one would expect the interactions to also affect the risk for disease relapse following allogeneic HSCT. This has, indeed, been found to be the case. In a retrospective study of a large cohort of patients who underwent allogeneic HSCT for various hematologic malignancies, intestinal presence of cluster of related Operational Taxonomic Units 1614 in the intestinal microbiome was associated with a lower risk for disease relapse. Further analysis found the correlation to be due primarily to an abundance of *Eubacterium limosum*.⁶³ This association was strongest among recipients of T-repleted grafts. The mechanisms for the apparent beneficial graft-versus-tumor effects of this anaerobic, nonspore forming Gram-positive rod remain speculative. Although it is possible that the change in the intestinal microbiota profile represents disease biomarker rather than being responsible for the reduced disease relapse, *E limosum* is known for its ability to produce short-chain fatty acids that possess immunomodulatory functions.⁶⁴ It is, therefore, likely that these short-chain fatty acids enhance the graft-versus-tumor effects following allogeneic HSCT.

5 | FMT RESTORES THE INTESTINAL MICROBIOME

FMT may be regarded as the pinnacle probiotic, given its currently unparalleled ability to restore any loss of diversity or change in the intestinal taxa or microbiome. It has a known role in refractory or relapsed CDI.⁶⁵ No other disease currently has FDA approved use of FMT.

Extending this therapy to allogeneic HSCT patients has been met with concerns of bacteremia and other associated infections which have been reported in immunocompetent patients.^{66,67} However, FMT has been successfully and safely carried out in patients who developed refractory CDI following allogeneic HSCT.⁶⁸ Six of the seven patients experienced long periods of remission of the CDI. A study is currently ongoing to evaluate the role of autologous FMT using stool sample collected prior to allogeneic HSCT to prevent post-transplant CDI (NCT02269150). The investigators in this study will also evaluate the development of intestinal GVHD in these patients to determine if restoration of the microbiome diversity using autologous FMT will reduce intestinal GVHD.

A recent study reported the application of FMT to treat refractory intestinal GVHD.⁶⁹ In this study of four patients with refractory intestinal GVHD, all four patients responded to the FMT, with three patients attaining a complete remission of the GVHD. Another study reported the successful use of repeated FMT in three patients who developed intestinal GVHD following allogeneic HSCT.⁷⁰ Obviously these results need to be confirmed in larger studies.

FMT has also been reported to also reduce intestinal colonization with antibiotic-resistant bacteria in patients with hematologic malignancies.⁷¹ It would, therefore, be of great interest to extend such an investigation to patients with a history of colonization with antibiotic-resistant bacteria prior to undergoing allogeneic HSCT to reduce the risk for the development of antibiotic-resistant infections.

There are currently a number of studies ongoing to investigate the role of FMT in preventing intestinal GVHD following allo-HSCT. One study investigates the feasibility of FMT after hematopoietic recovery after allogeneic HSCT and plans to evaluate the two-year overall survival, progression-free survival, as well as incidence of GVHD (NCT02733744). Taking this concept further, it might be worthwhile investigating the use of combined fecal microbiota and hematopoietic progenitor cells transplants from the same donor to reduce intestinal GVHD since the engrafted donor immune system will obviously be more tolerant to the donor intestinal microbiota than the host intestinal microbiota. The recent observations of the association between *E limosum* abundance in the intestinal microbiotic profile and reduced disease relapse also raises the possibility of modifying the intestinal microbiota of patients at high risk for disease relapse following allogeneic HSCT with *E limosum* inoculates to reduce the relapse rate.

6 | CONCLUSIONS

There exists mounting evidence promoting the considerable effect of the human intestinal microbiome on the clinical course following allogeneic HSCT. The anti-inflammatory mechanisms of certain species of commensals appear to be directly affected by alteration of the microbiome, whether by the natural changes of aging or after allogeneic HSCT or by iatrogenic means. The interaction between intestinal microbiotic shifts, mucosal inflammation and pathogenesis of GVHD may warrant more focused endeavors in the early detection of dysbiosis, which may include serum or even urinary markers for specific species (eg, *Enterococci*) whose alteration has been shown to be a potential surrogate for intestinal dysbiosis. Because of this possibility, an individualized microbiotic "stamp" may at some point prove to be part of the routine pretransplant evaluation. Beyond surveillance for such dysbiosis, we expect that there will be more directed efforts to alter the microbiome to improve the post-allogeneic HSCT outcome, whether by replacement with various probiotic organisms or preparations that lead to an increase in or colonization of specific commensal organisms.

SOURCE OF SUPPORT

None.

CONFLICT OF INTEREST

All authors declare no conflict of interest

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